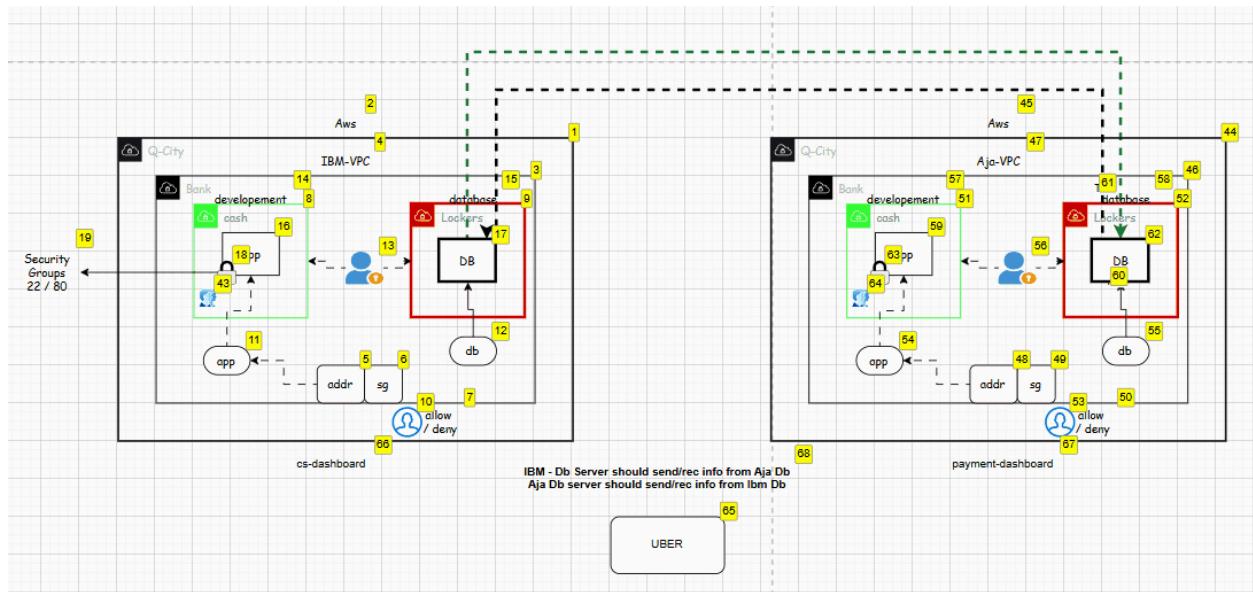




AWS VPC Peering – Complete Guide with Lab (Aja VPC ↔ Ibm VPC)

What is VPC Peering?

VPC Peering allows you to connect two VPCs privately so that resources (EC2, RDS, etc.) in one VPC can communicate with those in another as if they were on the same network, using private IPs.

Key Features

Feature	Description
 Bidirectional	Both VPCs can initiate traffic (once routes/Security Groups are set)

 Private	Data never leaves AWS backbone
 Cross-region	Supported (slightly more configuration)
 No Transitive Routing	VPC A ↔ VPC B works, but A can't route to C via B

Limitations

-  No transitive peering (must peer every pair directly)
 -  Cannot be renamed after creation
 -  Peering must be accepted by the other VPC
 -  Routes & security groups must be manually updated
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Lab: VPC Peering Between AjaVPC and IbmVPC

Goal

- Create 2 VPCs: **AjaVPC** and **IbmVPC**
 - Launch EC2 instances in each
 - Peer the two VPCs
 - Allow EC2 instances to ping each other using private IPs
-



6. Update Route Table to allow 0.0.0.0/0 via IGW

3 Launch EC2 Instances

VPC	Instance Name	Subnet	AMI	Security Group
AjaVPC	EC2-A	10.0.1.0/24	Amazon Linux 2	Allow SSH, ICMP
IbmVPC	EC2-B	192.168.1.0/24	Amazon Linux 2	Allow SSH, ICMP

Make sure both Security Groups allow inbound ICMP (ping) and SSH

4 Create VPC Peering Connection

1. Go to VPC → Peering Connections → Create Peering Connection
 2. Name: **Aja-to-Ibm**
 3. VPC Requester: **AjaVPC (10.0.0.0/16)**
 4. VPC Acceptor: **IbmVPC (192.168.0.0/16)**
 5. Click Create Peering
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5 Accept Peering Request

- Go to Peering Connections
- Select **Aja-to-Ibm**
- Click Actions → Accept Request

6 Update Route Tables

AjaVPC Route Table

- Destination: 192.168.0.0/16
- Target: Peering connection (pcx-xxxx)

IbmVPC Route Table

- Destination: 10.0.0.0/16
- Target: Peering connection (pcx-xxxx)

7 Update Security Groups

EC2-A SG (AjaVPC)

- Inbound Rule:
 - Type: ICMP (Echo Request)
 - Source: 192.168.0.0/16

EC2-B SG (IbmVPC)

- Inbound Rule:
 - Type: ICMP (Echo Request)
 - Source: 10.0.0.0/16

8 Test Connectivity

1. SSH into EC2-A

Run:

ping 192.168.1.100 # (Private IP of EC2-B)

- 2.
3. Expect ping replies! 🎉

Optional: Cross-Region Peering

- Peering can also be done between VPCs in different AWS regions
- Same steps, but you must ensure:
 - Subnets in different AZs
 - Security groups and route tables include regional peering

Summary Table

Step	Action
1	Create AjaVPC and IbmVPC
2	Launch EC2 in both
3	Create and Accept Peering
4	Update route tables
5	Allow ICMP in security groups

6 Ping from one EC2 to another
